

ToolMind: VLM-Guided Spatial Reasoning for Robotic Tool Use

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Abstract—We present a dual-modality pipeline for scenario-specific sequential robotic tool use. Given an RGB-D image of a tabletop scene, the system identifies the correct tool, predicts task-relevant grasp, functional contact, and waypoint keypoints, and executes a multi-phase manipulation sequence. A Vision-Language Model (VLM) handles semantic scene understanding, tool selection, and scenario-specific waypoint generation, while a depth-based geometric module refines coarse VLM outputs into precise 3D keypoints. A key challenge we address is *spatial myopia* of VLMs: their tendency to predict keypoints near the center of mass rather than at geometrically meaningful extremities. We mitigate this with a Structured Chain-of-Thought (CoT) JSON prompting strategy that forces step-by-step spatial reasoning before coordinate regression. Crucially, our approach requires no task-specific retraining of the VLM and operates in an open-world fashion: the same pipeline applies to arbitrary tabletop objects without scene-specific tuning. The resulting grasp, contact, and waypoint keypoints drive KOMO-based trajectory optimization in a simulated Franka Emika Panda environment across three distinct constrained scenarios: a wall-retrieval task, a screwdriver-insertion task, and a shelf-retrieval task. We evaluate the effect of Structured CoT prompting through end-to-end rollout ablations, focusing on whether the VLM selects functional contact points on the correct tool extremities.

Index Terms—robotic tool use, vision-language models, keypoint extraction, trajectory optimization, KOMO, chain-of-thought prompting, spatial grounding, point cloud manipulation

I. INTRODUCTION

Tool use is a fundamental hallmark of intelligent behaviour, yet robust autonomous tool use remains an open challenge in robotics. The difficulty arises from the need to simultaneously reason about semantics (*which tool is appropriate and why?*) and geometry (*where exactly should the robot grasp, and how should it move?*). Foundation models, and in particular Vision-Language Models (VLMs), have dramatically lowered the barrier to semantic scene understanding. VLMs can identify objects, infer affordances, and generate plausible action plans from natural language prompts without any task-specific fine-tuning—a property that makes them attractive for open-world deployment.

However, VLMs exhibit two related failure modes when tasked with precise spatial localisation for robotic tool use. First, even when the correct tool is identified, they tend to regress keypoint coordinates toward the centroid of the target object rather than to the geometrically meaningful extremity required for a given task, a phenomenon we term *spatial myopia*. Second, when VLMs are queried on raw scene

images without pre-segmented cluster labels, their predicted coordinates become substantially less grounded: they may fall on irrelevant objects, empty background regions, or scene locations unrelated to the selected tool. In other words, the absence of cluster-level grounding can cause the VLM not only to choose imprecise points on the tool, but also to produce points that are not physically associated with the tool at all.

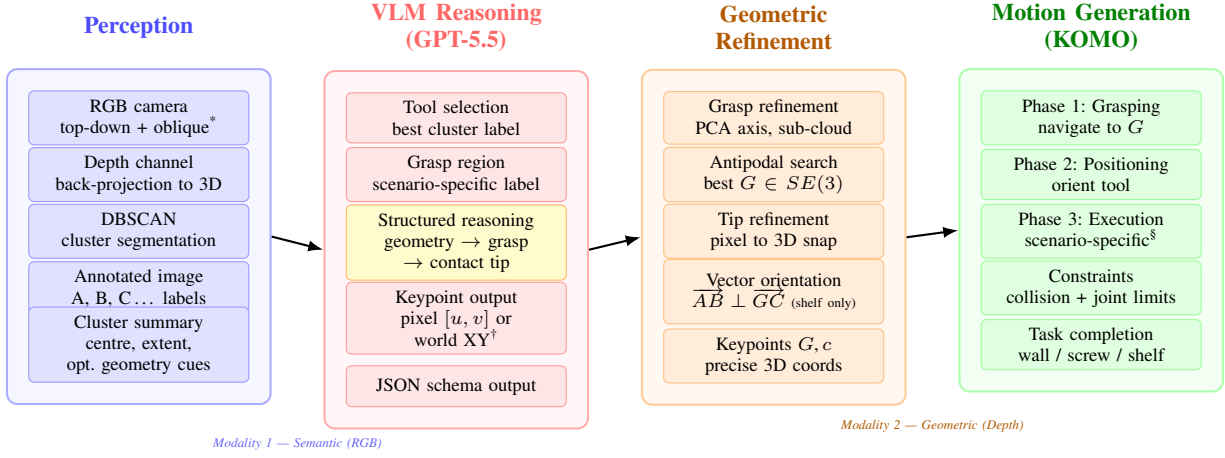
To address this, we provide the VLM with DBSCAN-derived cluster labels and per-cluster geometry summaries. This converts the problem from unconstrained image-level coordinate regression into a structured selection-then-localisation task, anchoring the model’s attention to candidate tool regions and reducing coordinate variance.

This paper addresses these grounding and spatial-myopia failures through a Structured CoT prompting strategy embedded within a strict JSON schema. Rather than asking the VLM for a pixel coordinate directly, we require it to first select a labelled cluster, reason about the tool geometry, identify the grasp region, and then infer the opposite functional contact point or scenario-specific waypoint. Our full pipeline combines this VLM reasoning front-end with a classical geometric back-end: depth-based point cloud extraction, DBSCAN clustering for scene segmentation, pixel-to-3D keypoint projection, and antipodal grasp search for physically feasible grasps.

II. RELATED WORK

Recent work has grounded language and vision-language models in robotic decision making. SayCan [1] maps language instructions to high-level robot skills using affordance scores, while Inner Monologue [2] uses language feedback to revise robot plans during execution. These approaches mainly operate at the task or skill-selection level. In contrast, our work targets the lower-level spatial grounding required for robotic tool use: selecting the correct tool, localising the grasp region, identifying the functional contact point, and generating scenario-specific waypoints.

Our pipeline also differs in how VLM outputs are used. Rather than directly executing raw VLM coordinates, we ground the model with DBSCAN-derived cluster labels and per-cluster geometry summaries. The VLM proposes semantically meaningful tool-use keypoints, while the geometric back-end refines them through depth projection, point-cloud snapping, antipodal grasp search, and KOMO-based execution. This combination allows the system to correct VLM spatial imprecision while preserving open-world semantic reasoning.



*Wall: primarily top-down (world-XY output). Shelf: top-down + oblique for geometry understanding. †Wall: world XY; shelf/screwdriver: pixel $[u, v]$. §Wall: pull. Screwdriver: rotate/insert. Shelf: sweep/push.

Fig. 1: Overview of the proposed pipeline across four stages.

III. SYSTEM ARCHITECTURE

The complete pipeline consists of four stages: (1) scene perception and point cloud generation, (2) VLM-based tool selection and structured CoT keypoint reasoning, (3) geometric keypoint refinement and antipodal grasp selection, and (4) KOMO-based trajectory optimisation and execution. Fig. 1 provides an overview of all stages.

A. Scene Perception and DBSCAN Clustering

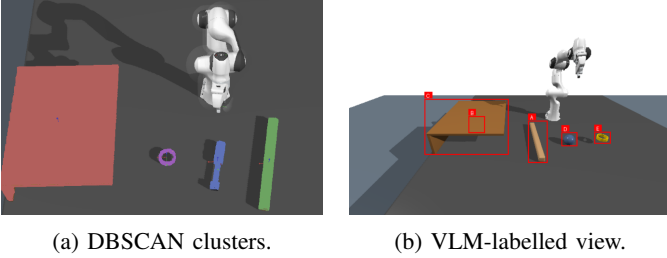


Fig. 2: Scene perception output. Movable object clusters are extracted from the RGB-D point cloud and projected back onto the RGB image as labelled VLM candidates.

At the start of each episode, RGB-D images are captured from calibrated top-down and oblique camera views. Depth pixels are back-projected using the camera intrinsics $\{f_x, f_y, c_x, c_y\}$ as

$$\mathbf{p}_{\text{cam}} = \left[\frac{(u - c_x)z}{f_x}, \frac{(v - c_y)z}{f_y}, z \right]^\top, \quad (1)$$

and transformed to world coordinates using the camera extrinsics $[\mathbf{R} \mid \mathbf{t}]$. The merged point cloud is segmented with DBSCAN, while static structures such as tables, walls, and shelf boards are removed using simple extent, elongation, and height heuristics. The remaining movable clusters are labelled A, B, C, ... and projected onto the RGB image for

VLM querying. These labels ground the VLM’s coordinate predictions to candidate tool regions rather than arbitrary image locations. An ablation without cluster labels confirms that this grounding improves spatial reliability by reducing high-variance predictions on irrelevant objects or background regions.

B. VLM-Based Tool Selection and Structured CoT Reasoning

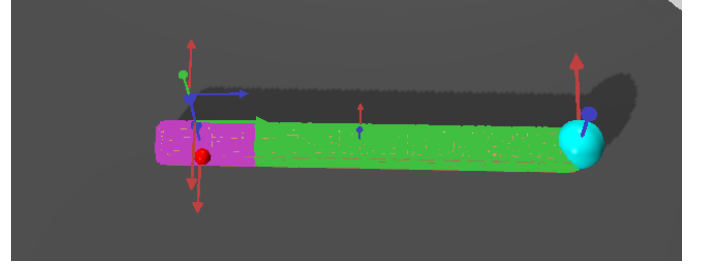


Fig. 3: VLM grasp/contact keypoints. The pixel (u, v) is lifted using average local depth and snapped to the selected tool cloud. Magenta shows the grasp region; cyan marks the functional contact point c .

The annotated RGB images and per-cluster geometry summaries (centroid, extent, elongation) are provided to GPT-5.5 through a structured JSON prompt.

To counteract spatial myopia, we use a constrained **Structured Chain-of-Thought Prompting** strategy. Before outputting coordinates, the VLM must first reason about the selected tool’s geometry, identify the grasp region, and infer the opposite functional contact point using the grasp location as an anchor.

Only after these reasoning steps does the schema request *functional_contact_pixel* and *contact_point_rationale*. This forces the model to reason about geometric extremities before

coordinate output, reducing the tendency to place keypoints near the object centre or at non-functional regions.

C. VLM Pixel-to-3D Projection

The VLM returns a pixel coordinate (u, v) for each predicted grasp, contact, or waypoint keypoint. Valid depth values are averaged in a small region around (u, v) and used in Eq. (1) to lift the prediction to 3D. The resulting point is transformed to the world frame, snapped to the selected tool cloud when the keypoint lies on the tool, and used to define a 4 cm crop for antipodal grasp search. This yields the final gripper pose $G \in SE(3)$ and approach direction.

D. VLM Waypoint Generation

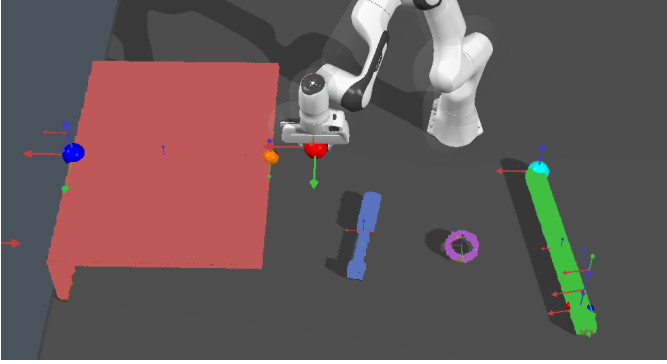


Fig. 4: VLM waypoint generation. Markers show wall pull points, screwdriver screw-centre offset, and shelf-edge anchors A and B.

Beyond the grasp keypoint, each scenario requires auxiliary waypoints that guide the multi-phase trajectory (Fig. 4). All waypoints are returned as pixel coordinates by the VLM and projected to 3D via the same back-projection pipeline of Eq. (1).

Wall-pull waypoints. The VLM identifies two waypoints: one *behind the target box*, where the tool tip hooks the box from the far side of the wall, and one *between the wall and the box* as the pull endpoint. These points define the pull vector and serve as sequential KOMO phase goals.

Screwdriver waypoint. The VLM identifies the screw centre from the annotated image. A fixed offset *above* this point along the screw axis converts the 2D prediction into a collision-safe 3D descent pose before rotation.

Shelf waypoints. The VLM identifies two points A and B located on the two opposite long edges of the shelf gap (one on the robot-near edge, one on the far edge). These anchor points define the sweep direction vector $\vec{AB}$ and are used exclusively for tool orientation alignment (Section III-E); the actual sweep trajectory is governed by the grasp–contact keypoints G and c .

E. Vector-Based Tool Orientation (Shelf Scenario)

In the shelf scenario the tool must be inserted into a narrow gap without colliding with the shelf boards, regardless of the

tool’s initial placement in the scene. The VLM is additionally queried for two points A and B located on the two opposite long edges of the shelf gap (one on the robot-near side, one on the far side), defining the sweep direction vector $\vec{AB}$. Simultaneously, the grasp keypoint G and the functional contact point C on the tool define the tool axis vector $\vec{GC}$. The gripper frame is then rotated until the two vectors are perpendicular:

$$\theta = \arccos\left(\frac{\vec{AB} \cdot \vec{GC}}{\|\vec{AB}\| \|\vec{GC}\|}\right) = 90^\circ, \quad (2)$$

ensuring that the tool always enters the shelf gap aligned with the sweep direction, irrespective of its initial world-frame orientation. This vector-based strategy is fully VLM-agnostic in the geometric computation: the VLM contributes only the semantic identification of the shelf edges and the tool endpoints; the orientation correction is computed analytically.

F. KOMO Trajectory Optimisation

The refined keypoints G and c drive a multi-phase KOMO [3] trajectory in the RAI framework with a Franka Emika Panda robot. Each sequence consists of three phases: (1) *Grasping*—navigate to G and close the gripper; (2) *Positioning*—reorient the tool to the scenario-appropriate pose; (3) *Execution*—drive tip c toward the target using a push, pull, hook, or screw motion. Each phase includes joint-limit constraints and collision avoidance.

IV. EXPERIMENTAL SETUP

A. Scenarios

We evaluate the pipeline on three constrained tabletop scenarios, each with three scene variants that randomise tool pose, orientation, or cluster labels to prevent label-copying artefacts.

TABLE I: Evaluation scenarios.

Scenario	Task	Variation
Wall	Rake pulls a box from behind a wall into reach	Rake pose/orientation
Screwdriver	Screwdriver tip enters a rotating target	Tool pose/cluster label
Shelf	Sweeping tool moves a cup out from under a shelf	Tool pose/cluster label

V. RESULTS

A. Prompt and Grounding Ablation

Before the end-to-end CoT rollout study, we evaluate how prompt detail, visual cluster grounding, and model size affect VLM keypoint quality. Annotated mode uses DBSCAN-derived bounding boxes and labels, while raw mode removes these anchors and requires the VLM to reason from unlabelled objects. P0 is minimal, P1 adds task and grasp rules, P2 adds object geometry and affordance hints, and P3 adds explicit anti-failure constraints.

TABLE II: Wall retrieval ablation. Hit = 2D grasp hit rate; AFR = antipodal feasibility rate; Dev. = grasp deviation in mm.

Model	Mode	Prompt	Tool	Hit	AFR	Dev.
4o	Ann.	P0	0.57	0.00	0.07	210.2
4o	Ann.	P1	0.67	0.00	0.08	231.5
4o	Ann.	P2	0.73	0.13	0.20	160.6
4o	Ann.	P3	0.93	0.87	0.87	32.8
4o	Raw	P0	0.73	0.00	0.00	220.4
4o	Raw	P1	0.72	0.29	0.29	220.3
4o	Raw	P2	0.87	0.00	0.00	245.7
4o	Raw	P3	1.00	1.00	0.93	17.6
4o-mini	Ann.	P0	0.90	0.20	0.07	119.1
4o-mini	Ann.	P1	0.70	0.20	0.00	131.9
4o-mini	Ann.	P2	0.77	0.07	0.07	185.2
4o-mini	Ann.	P3	0.95	0.73	0.33	7.1

TABLE III: Screwdriver insertion ablation. Hit = 2D grasp hit rate; AFR = antipodal feasibility rate; Dev. = grasp deviation in mm.

Model	Mode	Prompt	Tool	Hit	AFR	Dev.
4o	Ann.	P0	0.70	0.10	0.20	47969.0
4o	Ann.	P1	0.67	0.10	0.65	71.2
4o	Ann.	P2	0.80	0.67	0.65	1.1
4o	Ann.	P3	0.80	0.83	0.65	0.2
4o	Raw	P0	0.52	0.03	0.13	27037.0
4o	Raw	P1	0.50	0.15	0.20	215.7
4o	Raw	P2	0.70	0.46	0.37	37.4
4o	Raw	P3	0.71	0.60	0.49	4.5
4o-mini	Raw	P0	0.06	0.04	0.20	3252.4
4o-mini	Raw	P1	0.56	0.30	0.37	48.7
4o-mini	Raw	P2	0.82	0.54	0.68	16.1
4o-mini	Raw	P3	0.74	0.72	0.69	8.0
4o-mini	Ann.	P0	0.72	0.17	0.43	1652.0
4o-mini	Ann.	P1	0.69	0.32	0.49	38.3
4o-mini	Ann.	P2	0.74	0.67	0.56	8.5
4o-mini	Ann.	P3	0.81	0.83	0.77	0.7

Table II shows that wall retrieval strongly benefits from explicit anti-failure guidance. P3 gives the highest hit rate and the lowest deviation in every reported setting. We therefore treat P3 as the most reliable wall prompt, while noting that raw-mode gains partly reflect stronger prompt constraints rather than purely visual grounding.

For screwdriver insertion, the main trend is progressive stabilization from P0 to P3. Minimal prompts can produce extremely large deviations, especially without strong spatial grounding. P2 and P3 substantially reduce this variance, and P3 gives the best or tied-best hit rate in the 4o annotated and 4o-mini annotated settings.

Shelf retrieval remains the hardest perception setting because the VLM must identify both a graspable tool region and the correct functional end for sweeping. P3 improves tool selection and reduces deviation across all reported shelf settings, but the hit and AFR values show that localizing a physically useful contact geometry is still less stable than in the wall or screwdriver tasks. This motivates the end-to-end CoT analysis below, where the main remaining failure mode is whether C is placed on the correct opposite tool extremity.

TABLE IV: Shelf retrieval ablation. Hit = 2D grasp hit rate; AFR = antipodal feasibility rate; Dev. = grasp deviation in mm.

Model	Mode	Prompt	Tool	Hit	AFR	Dev.
4o	Ann.	P0	0.42	0.25	0.07	136.7
4o	Ann.	P1	0.65	0.30	0.05	147.7
4o	Ann.	P2	0.70	0.05	0.07	186.5
4o	Ann.	P3	0.90	0.65	0.07	56.1
4o	Raw	P0	0.62	0.15	0.04	112.1
4o	Raw	P1	0.70	0.20	0.02	132.1
4o	Raw	P2	0.70	0.05	0.07	186.5
4o	Raw	P3	0.90	0.55	0.06	56.1
4o-mini	Ann.	P0	0.78	0.20	0.05	125.1
4o-mini	Ann.	P1	0.76	0.30	0.00	149.1
4o-mini	Ann.	P2	0.79	0.07	0.07	191.5
4o-mini	Ann.	P3	0.90	0.55	0.27	28.1

TABLE V: End-to-End Rollout Success: CoT vs. No-CoT (10 rollouts per scenario)

Scenario	With CoT	Without CoT
Wall retrieval	8 / 10	7 / 10
Screwdriver	7 / 10	7 / 10
Shelf retrieval	6 / 10	3 / 10

B. End-to-End CoT Rollout Ablation

We evaluate the effect of Structured CoT prompting through full task rollouts across all three scenarios. Table V summarizes the end-to-end success rates with and without CoT.

The strongest effect appears in the shelf retrieval task, where the VLM must reason about the shelf-gap orientation, the tool axis, and the correct functional end of the tool. With Structured CoT, 6 out of 10 shelf rollouts succeeded in sweeping the cup out from under the shelf, compared with 3 out of 10 without CoT. The failure breakdown in Table VI shows that the improvement is mainly due to better functional contact point localization.

Structured CoT reduces C-placement failures from 5 to 2 while leaving the A–B edge failure mode unchanged. This isolates the benefit of CoT to functional contact-point localization: the model more often places C on the opposite functional extremity of the tool rather than near the grasp point or tool centre. The improvement is smaller in the wall scenario and absent in the screwdriver scenario, suggesting that CoT is most useful when the task requires relational geometry over multiple keypoints.

VI. DISCUSSION

Open-world generalisation. The pipeline requires no re-training or scene-specific tuning of the VLM. DBSCAN parameters are the only per-scenario hyper-parameters, and these depend only on coarse sensor characteristics (camera height, object scale) rather than object identity. Consequently, the same pipeline handles arbitrary tool-object combinations, making it suitable for unstructured, open-world environments such as factory floors or domestic service settings.

TABLE VI: Shelf Failure Breakdown (10 rollouts)

	With CoT	Without CoT
Successful rollouts	6 / 10	3 / 10
<i>Failure breakdown</i>		
Wrong A–B edge pair	2	2
C placed at same end as grasp	2	5
Total failures	4	7

The cluster-label effect. A key design choice that emerged from our experiments is the importance of supplying DBSCAN cluster labels to the VLM. In preliminary trials without labels, the VLM was queried on the raw annotated image without per-cluster geometry summaries; predicted pixel coordinates showed high inter-run variance and frequently landed on background or on the wrong object. Adding cluster labels reduces this variance by anchoring the VLM’s spatial reasoning to discrete, labelled regions—converting an unconstrained image regression into a structured selection-then-localisation problem.

Limitations. Although cluster labels and Structured CoT improve spatial grounding, the system still depends on the VLM’s ability to identify task-relevant geometric relations. In the shelf scenario, failures can still occur when the VLM selects the wrong A–B shelf-edge pair or places the functional contact point C on the same end as the grasp point rather than the opposite tool extremity. These errors indicate that relational spatial reasoning remains challenging even with structured prompting.

The pipeline also relies on accurate depth reconstruction and segmentation. Noisy depth near object boundaries, missing point-cloud regions, or incorrect DBSCAN clusters can lead to inaccurate 3D keypoints after projection and snapping. Finally, execution is mostly open-loop: once the KOMO trajectory is generated, the system does not re-plan if the tool slips, contacts the object unexpectedly, or fails to move the target as intended. Future work should add closed-loop visual feedback and recovery behaviours during manipulation.

VII. CONCLUSION

We presented a dual-modality pipeline for scenario-specific sequential robotic tool use that couples VLM semantic reasoning with depth-based geometric refinement and KOMO trajectory optimisation. The central contribution is a Structured CoT prompting strategy that measurably mitigates VLM spatial myopia, improving functional keypoint localisation purely through prompt engineering. End-to-end CoT ablations show that explicit spatial reasoning improves the most geometrically constrained task, increasing shelf retrieval success from 3/10 to 6/10 by reducing functional contact-point failures. The system requires no VLM fine-tuning and remains applicable to arbitrary open-world tabletop scenes.

TASK SHARE

Süleyman Talha Belen: General pipeline design; VLM grasp point and waypoint generation; 3D keypoint mapping from VLM outputs; CoT prompting strategy design for task-specific scenarios.

Yusuf Buğra Yazıcıoğlu: Point cloud generation for each scene; object clustering via DBSCAN.

Yunus Emre Kök: KOMO trajectory execution for each scenario.

Mehmet Emin Yakşı: Scenario design and environment setup (wall retrieval, screwdriver insertion, shelf retrieval).

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